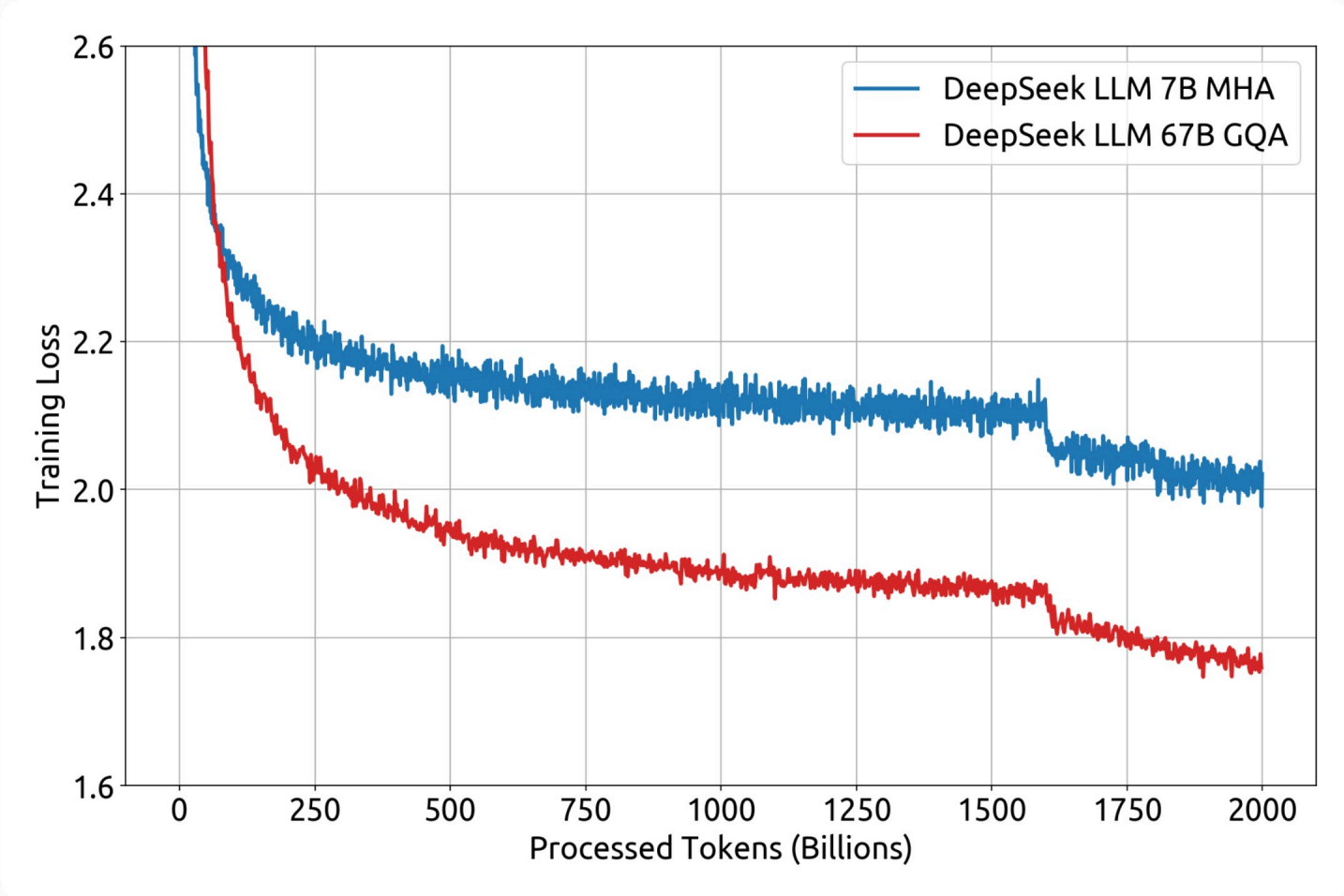
Same Shape at Frontier Scale

2 trillion

tokens of text

DeepSeek LLM 7B and 67B (DeepSeek-AI, 2023)



Our Real Model: Pythia-410M

410 million
weights

300 billion
tokens of training text
(trained by EleutherAI)

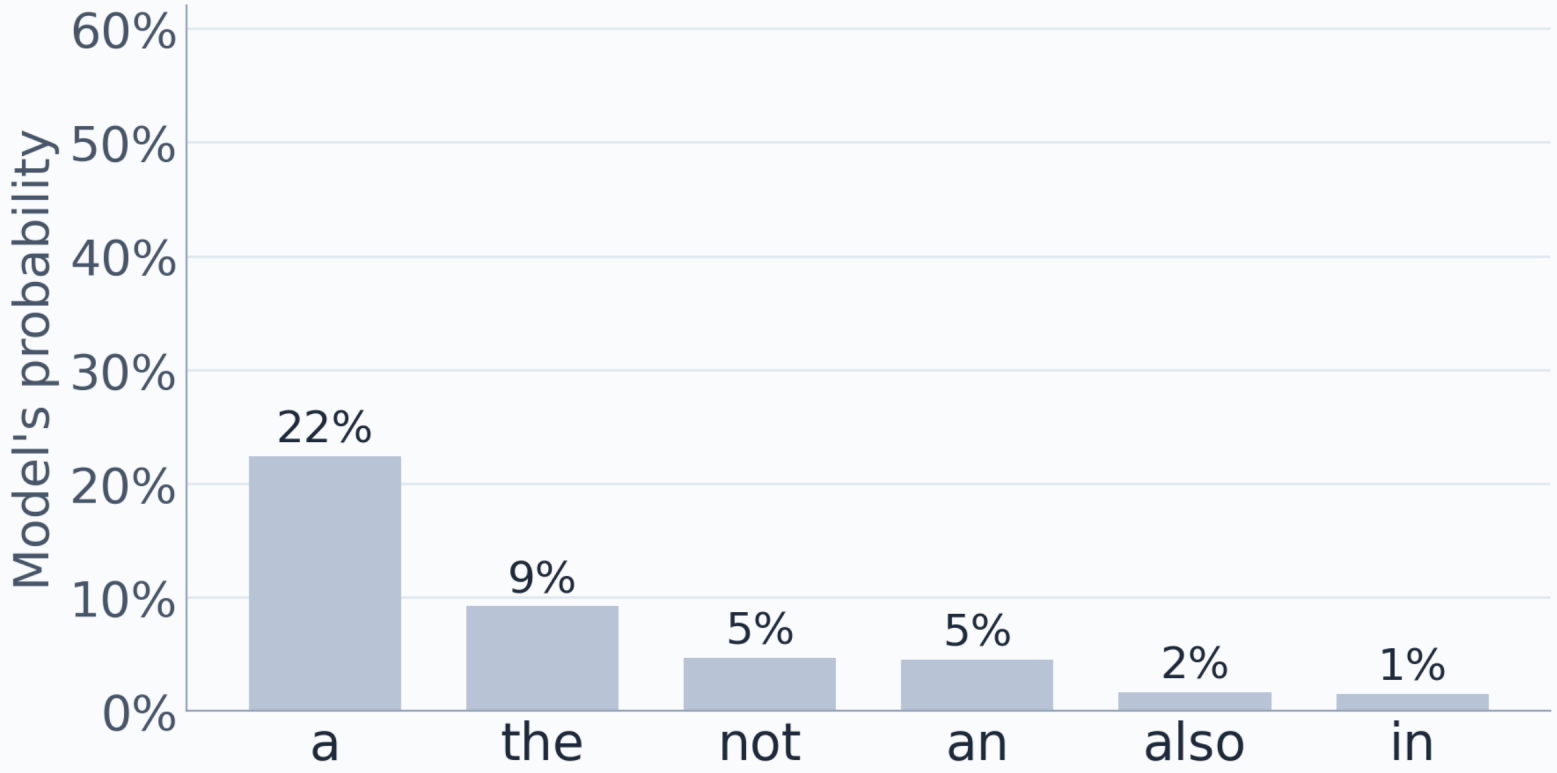
154
snapshots saved
during training

We downloaded 13 snapshots to a laptop and asked each one to finish:

> The capital of France is Paris. The capital of Japan is Tokyo. The capital of Texas is ___

Watching a Real Model Learn

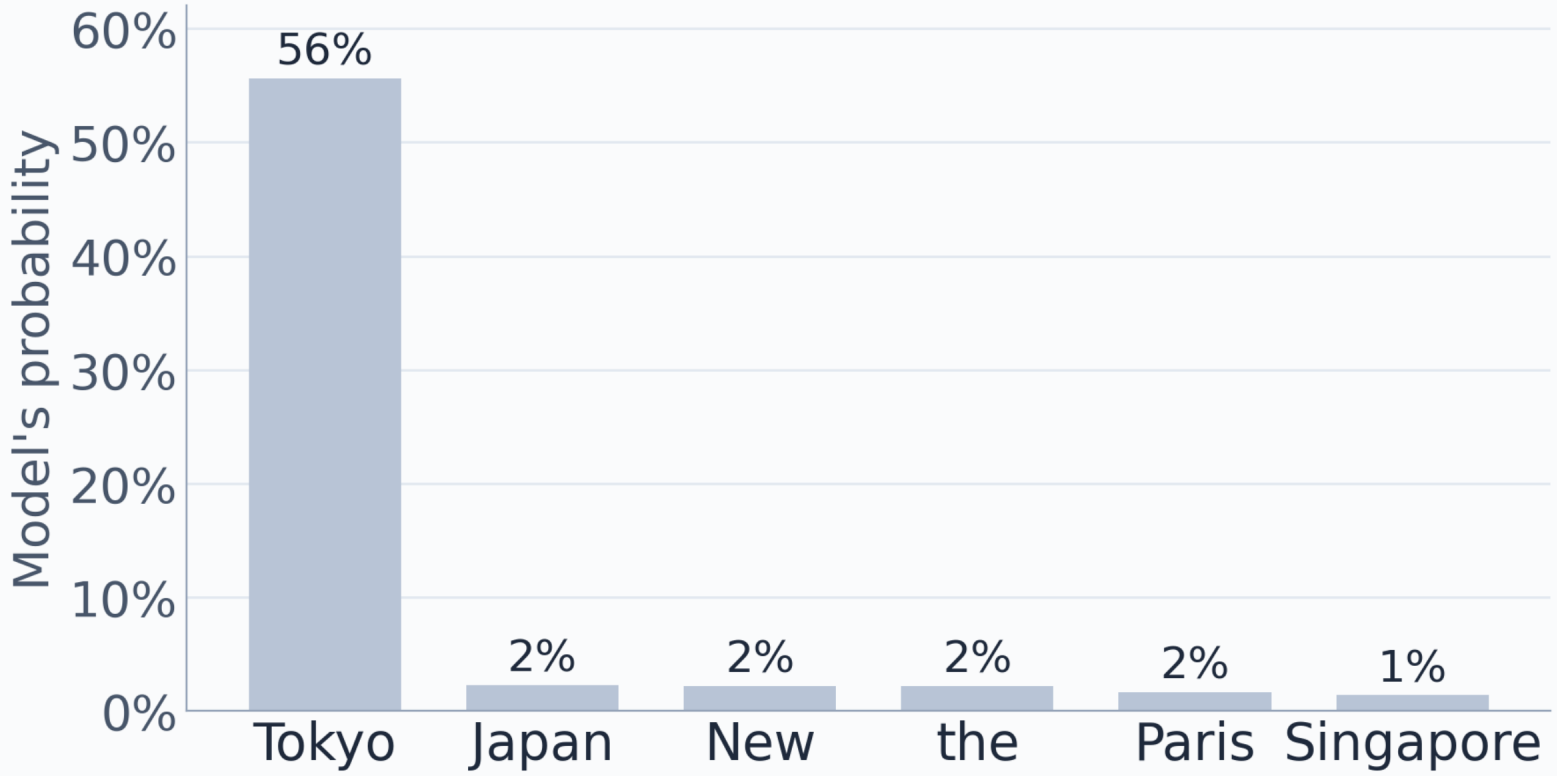
The capital of France is Paris. The capital of Japan is Tokyo. The capital of Texas is ___



Step 512

Watching a Real Model Learn

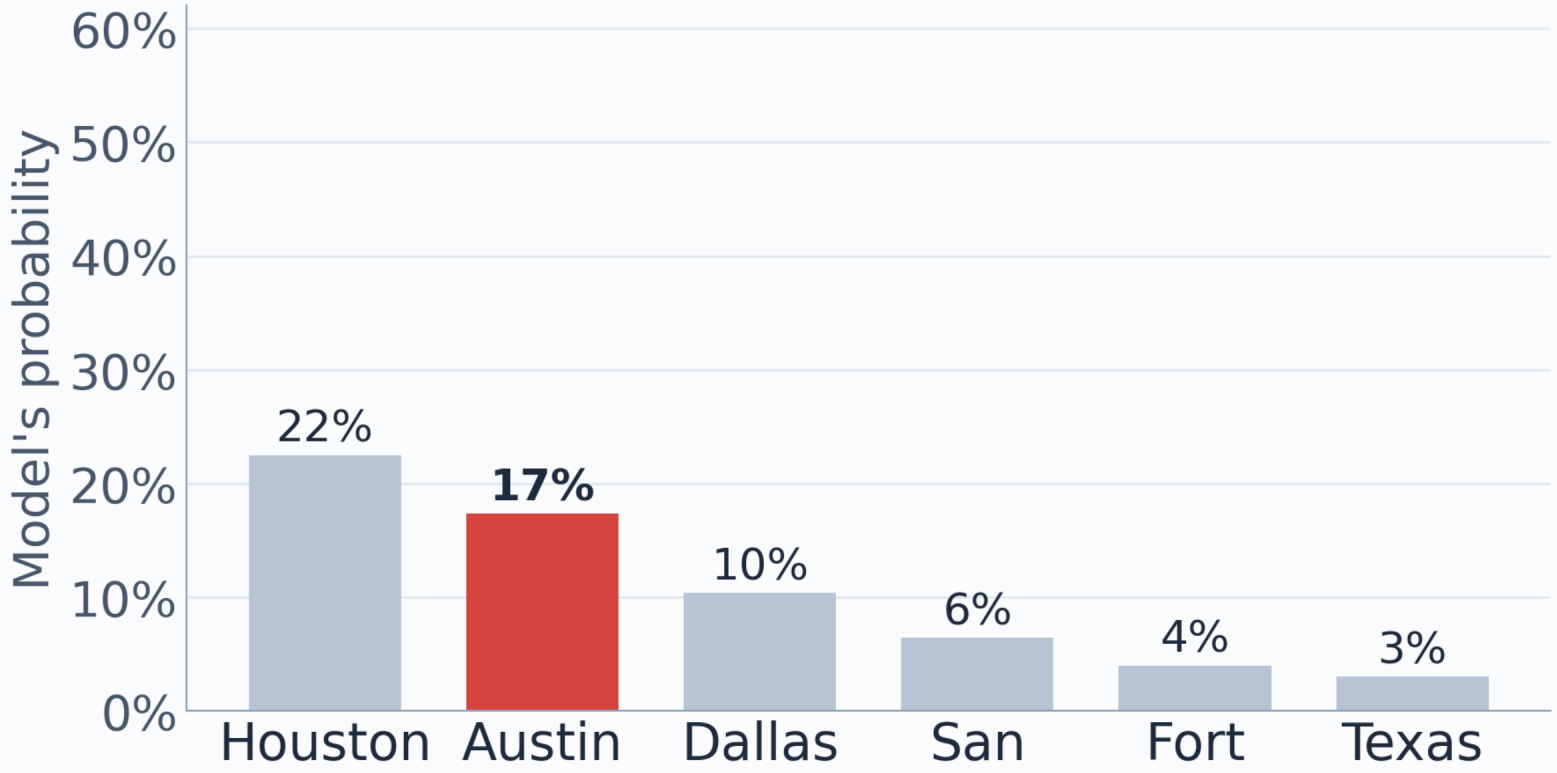
The capital of France is Paris. The capital of Japan is Tokyo. The capital of Texas is ___



Step 2,000

Watching a Real Model Learn

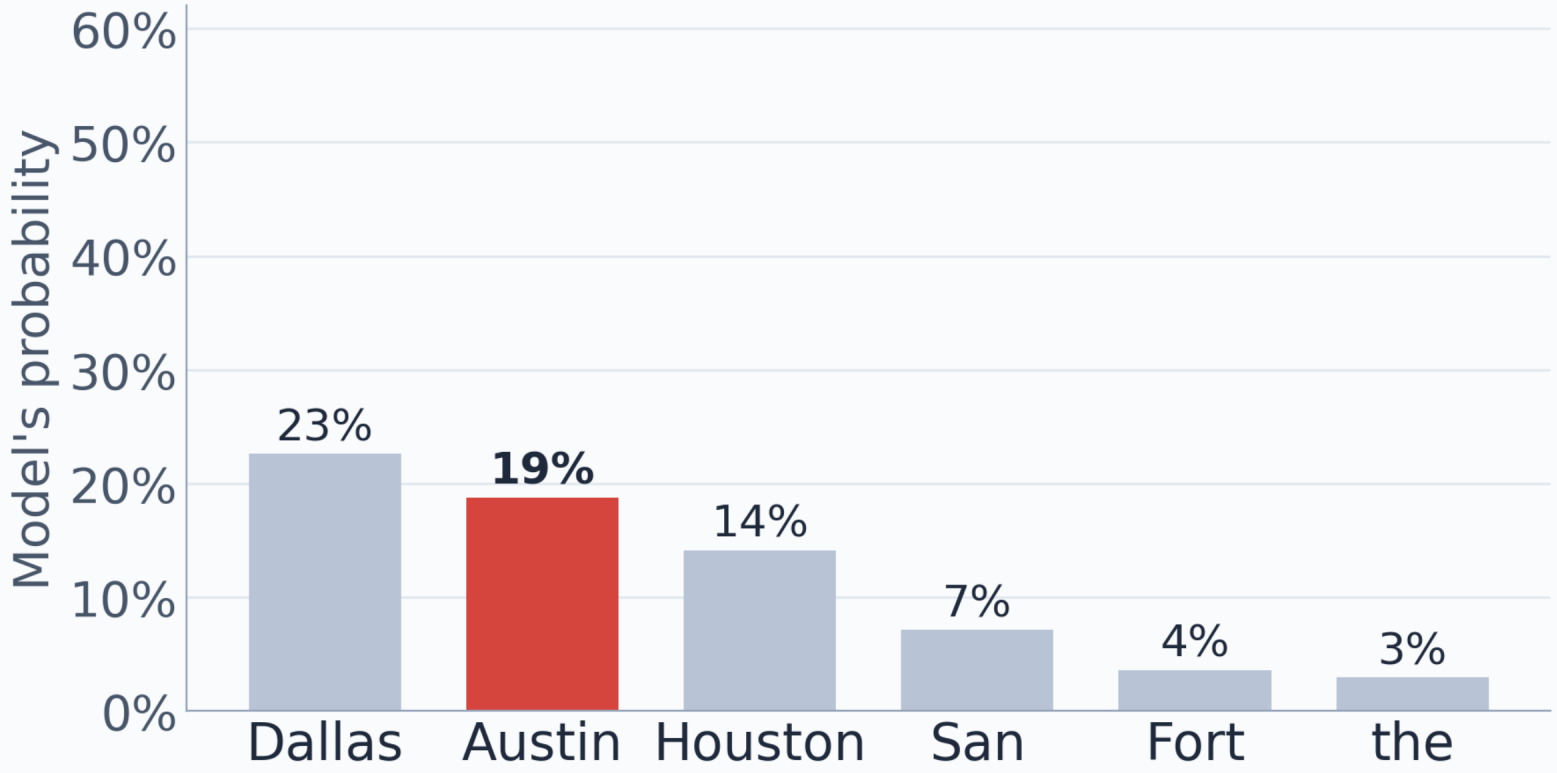
The capital of France is Paris. The capital of Japan is Tokyo. The capital of Texas is ___



Step 8,000

Watching a Real Model Learn

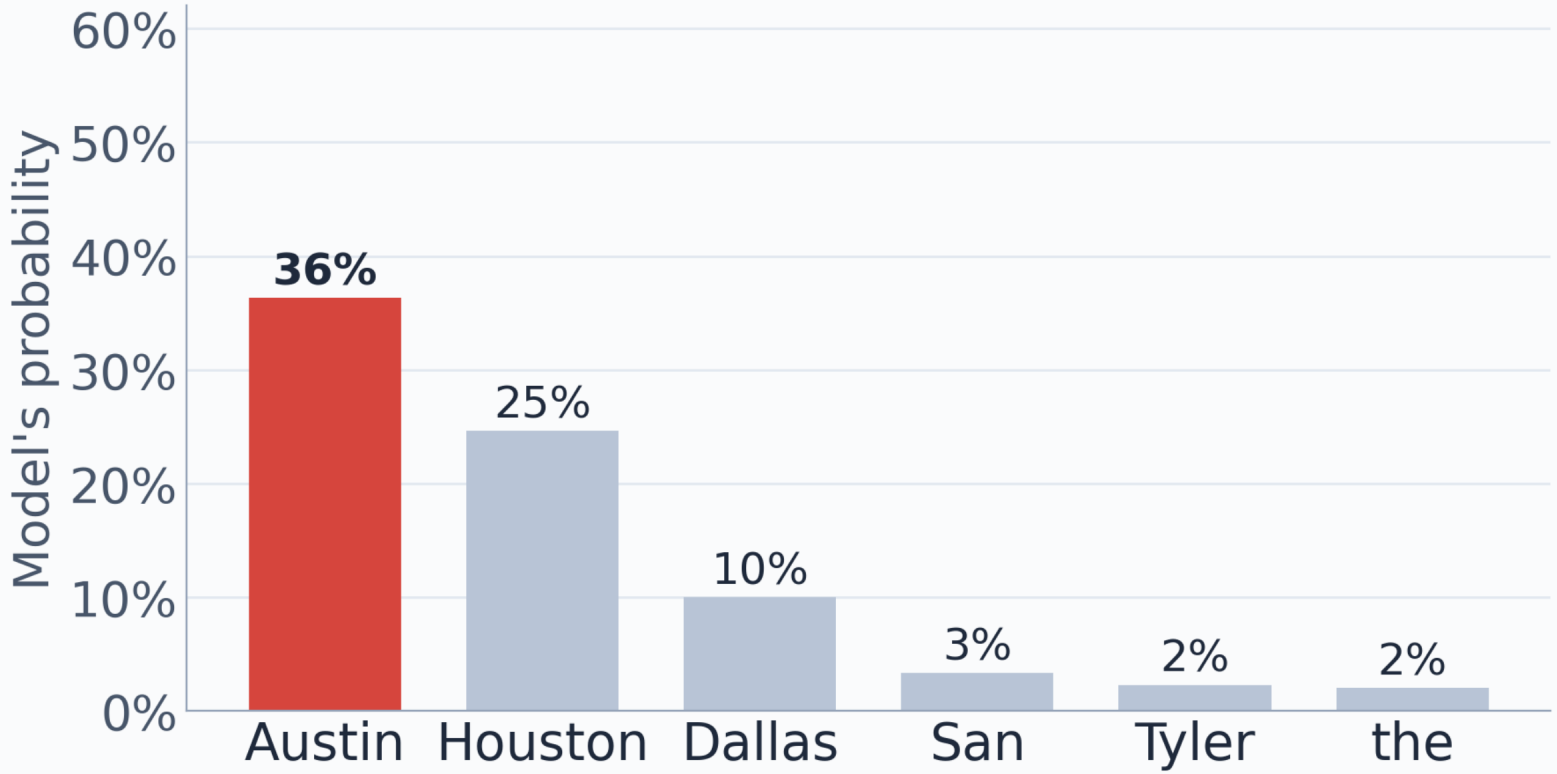
The capital of France is Paris. The capital of Japan is Tokyo. The capital of Texas is ___



Step 16,000

Watching a Real Model Learn

The capital of France is Paris. The capital of Japan is Tokyo. The capital of Texas is ___

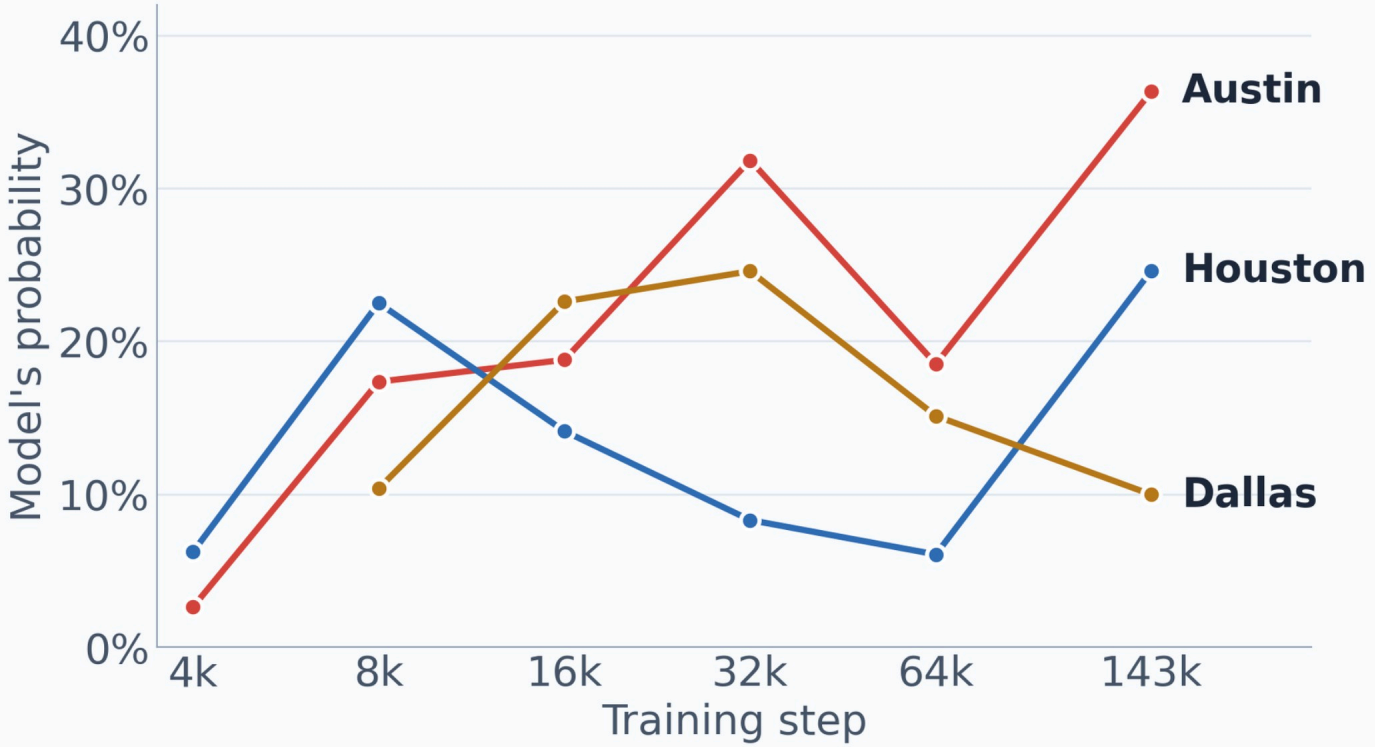


Step 143,000

final model

Sentences Pull in Different Directions

The model learns how often things are said, **not what is true**.



“Once upon a time, ...”

Step 0	ChicagoeachYour Handlekeepers descendedstoryPaint Birmingham...
Step 256	one or "The value, it is a bit of various reason, and would have been a, and it arranged, the death...
Step 2,000	one or two of the people in whom the government is charged may be charged with the duty of the court.
Step 143,000	one day, we saw a water bottle," he said. "I saw a bottle and I grabbed it. And I was like, 'Oh my...

Pretraining: Quality over Quantity

Reddit data

$60M / year

paid by Google

Physical books

Millions

bought and scanned by Anthropic